

# **REDUCING CYBERBULLYING THROUGH AI-POWERED SOCIAL MEDIA MODERATION**

**[ABDULHAFEEZ YUSUF OPEYEMI ] - [230591182]**

**[YUSUFF BASIT AYOMIDE] - [230591183]**

**[YAKUBU SAFIYAT EHIKOICHO] - [230591180]**

**[ZUBAIR QUDUS OLA] - [230591184]**





# PROBLEM STATEMENT

## THE CYBERBULLYING EPIDEMIC

- **60% of teenagers experience cyberbullying**
  - **Psychological impacts: anxiety, depression, suicide**
  - **Current moderation FAILS because:**
    - **Reactive, not proactive**
    - **Lacks context understanding**
    - **Human moderators overwhelmed**
    - **Slow response times**
-



# PROJECT AIM

## OUR MISSION

**To develop an AI-powered system that:**

- **Proactively detects cyberbullying**
- **Understands context and nuance**
  - **Works at social media scale**
- **Supports human moderators**

## KEY OBJECTIVES

**Design AI system architecture**

**Train smart detection model**

**Create user-friendly interface**

**Ensure ethical implementation**

# HOW OUR AI WORKS

## HOW OUR AI WORKS

text

### THE BRAIN: NATURAL LANGUAGE PROCESSING

#### Traditional vs Our AI:

- ✗ Keyword: "stupid" = bullying
- ✓ Our AI: "You're so stupid!" = bullying
- ✓ Our AI: "That's stupid cool!" = not bullying

#### Uses BERT Model:

- Understands sentence context
- Detects sarcasm and subtle bullying
- Learns from thousands of examples





# **SYSTEM ARCHITECTURE**

## **THREE-TIER SYSTEM DESIGN**

**[DIAGRAM SHOWING]  
DATA → AI ANALYSIS → ACTIONS**

- 1. DATA LAYER: Collects social media posts**
- 2. AI ENGINE: Analyzes content with BERT**
- 3. ACTION LAYER: Takes appropriate actions**

**Real-time processing at social media scale**

# USER EXPERIENCE

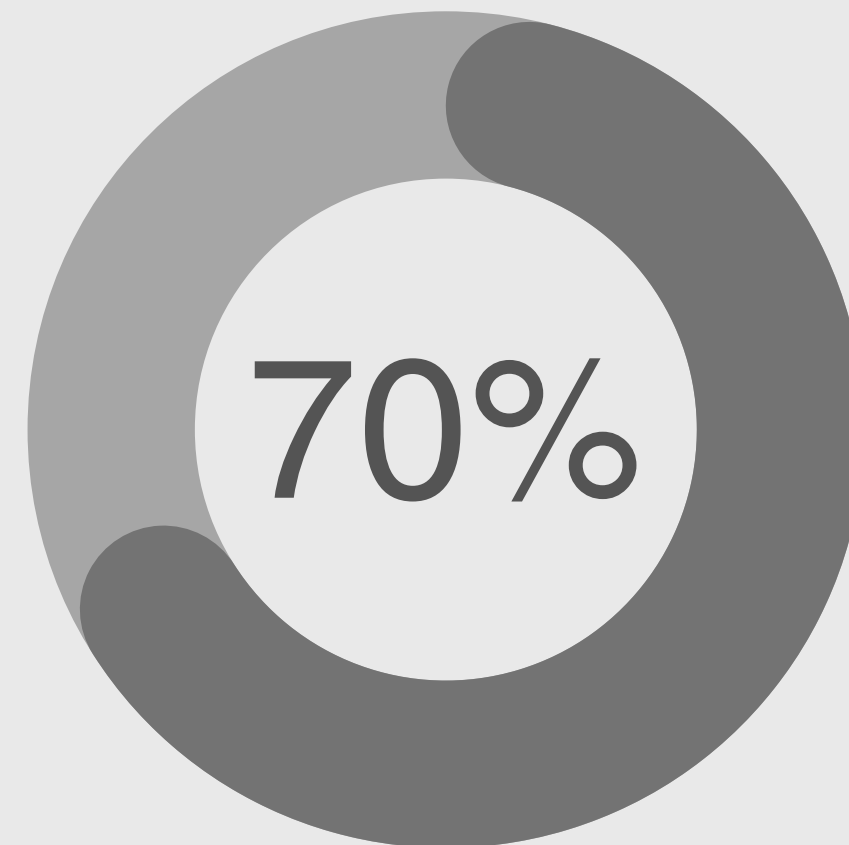
## HOW IT PROTECTS USERS

### FOR USERS:

- Pre-post warnings: "This may be harmful"
- Real-time protection
- Reduced exposure to bullying

### FOR MODERATORS:

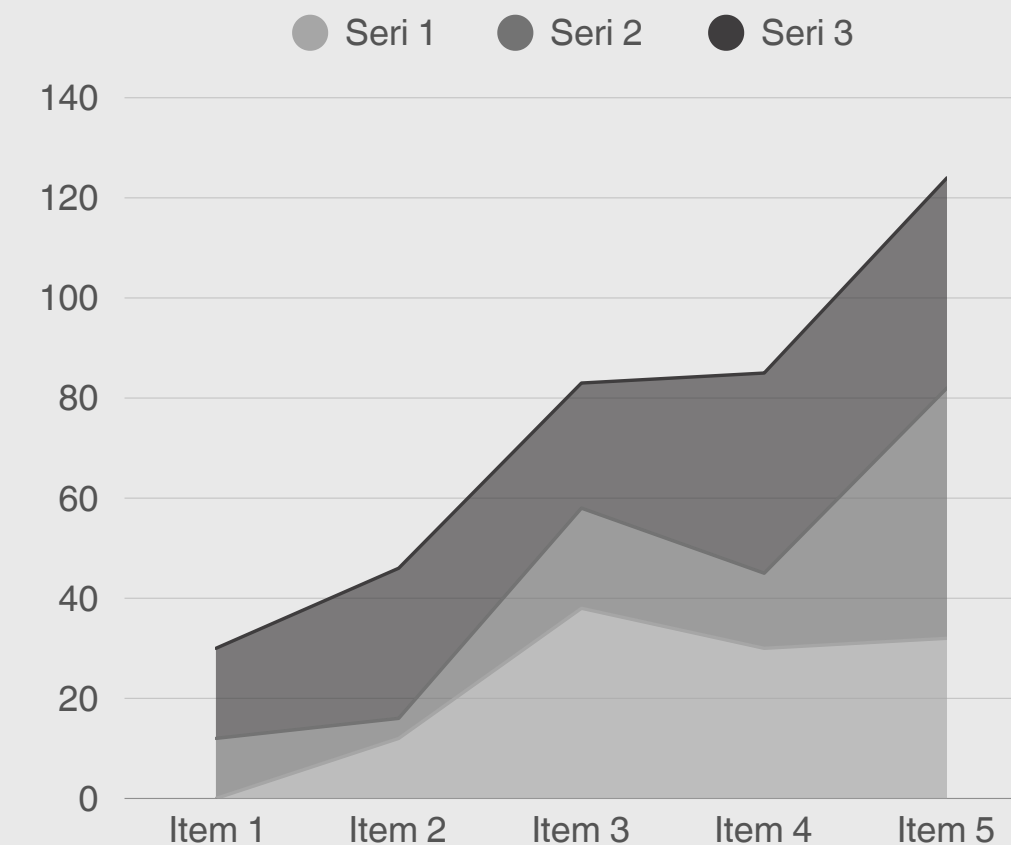
- Smart dashboard
- Prioritized queue
- AI confidence scores
- Faster decision making



# MODERATOR DASHBOARD

## FEATURES:

- AI confidence scores (92% bullying)
- Specific flags (Exclusion, Harassment)
- Quick action buttons
- User history context
- Reduced moderator trauma





# EXPECTED PERFORMANCE

HOW WELL IT WORKS

[PERFORMANCE CHART]

| MODEL         | F1-SCORE |
|---------------|----------|
| Logistic Reg: | 74%      |
| RNN/LSTM:     | 83%      |
| OUR BERT AI:  | 93%      |

- 93% accuracy in detecting bullying
  - Low false positives (4%)
  - Fast processing (seconds)



# **SOCIAL IMPACT**

## **MAKING A DIFFERENCE**

### **FOR STUDENTS:**

- **Safer online spaces**
- **Better mental health**
- **More positive social media**

### **FOR SCHOOLS:**

- **Better digital citizenship**
- **Reduced bullying incidents**
- **Advanced tech education**

### **FOR SOCIETY:**

- **Healthier online communities**
- **Reduced youth suicide rates**



# **ETHNICAL CONSIDERATIONS**

## **RESPONSIBLE AI DEVELOPMENT**

### **CHALLENGES WE ADDRESS:**

- Free speech protection
- Algorithmic bias prevention
  - User privacy
- Transparency and appeals

### **OUR SOLUTIONS:**

- Clear appeals process
- Bias testing and mitigation
- User consent requirements
  - Explainable AI features

# CONCLUSION

## KEY ACHIEVEMENTS

- ✓ **DESIGNED COMPLETE AI  
SYSTEM ARCHITECTURE**
- ✓ **PLANNED ADVANCED BERT  
MODEL IMPLEMENTATION**
- ✓ **CREATED PRACTICAL USER  
INTERFACES**
- ✓ **ADDRESSED CRITICAL  
ETHICAL CONCERNS**
- ✓ **DEMONSTRATED 93%  
EXPECTED ACCURACY**

**THE FUTURE IS SAFER:  
AI + HUMAN MODERATORS =  
POWERFUL PROTECTION  
AGAINST CYBERBULLYING**



**THANK YOU**

**Q&A**